Synoptic Question 11: The Physics of a Formula 1 Racing Car

Reading Passage

Read the following article and then answer the questions that follow.

A Formula 1 (F1) car is an extreme example of automotive engineering, designed to achieve the fastest possible lap times. This requires not only a powerful engine but also incredibly high cornering speeds and braking capabilities.

A modern F1 car uses a hybrid V6 turbo engine, which combines a traditional internal combustion engine (ICE) with an Energy Recovery System (ERS). The ERS includes a Motor Generator Unit-Kinetic (MGU-K), which acts as a generator during braking. It converts some of the car's kinetic energy, which would otherwise be lost as heat in the brakes, into electrical energy. This energy is stored in a high-voltage battery. During acceleration, the MGU-K reverses its function, acting as a motor to deliver additional power to the wheels.

Cornering performance is dominated by aerodynamics. The car's wings and bodywork are designed to create a powerful downward aerodynamic force, known as "downforce". This force, which increases with the square of the car's speed, presses the car onto the track, dramatically increasing the normal force on the tyres. This allows for a much greater frictional force between the tyres and the track, enabling the car to take corners at speeds that would be impossible for a conventional vehicle.

The sound of an F1 car is also distinctive. As the car passes a spectator at high speed, the observed pitch of the engine sound changes dramatically due to the Doppler effect.

Data:

Data for a typical F1 car:

Table 1: Data for Formula 1 Car

Parameter	**Value**
Total mass of car and driver, m	800 kg
Time for 0 to 100 km/h acceleration	2.6 s
Maximum power of internal combustion engine (ICE)	750 kW
Maximum speed on a long straight, v_{top}	350 km/h
Speed during a high-speed corner, v_c	180 km/h
Radius of this corner, r	80 m
Aerodynamic downforce at 180 km/h, F_{down}	15000 N
Coefficient of static friction for tyres, μ_s	1.5
Braking speed range for MGU-K	300 km/h to 100 km/h
Time taken for this braking, t_{brake}	2.0 s
Efficiency of MGU-K energy recovery, η_K	60%
True frequency of engine sound, f_s	800 Hz
Speed of sound in air, v_{sound}	343 m s^{-1}

Questions

(a) During the start of a race, the car accelerates from 0 to 100 km/h in 2.6 s.

  (i) Convert 100 km/h to m s^{-1}.

  (ii) Assuming constant acceleration, calculate this acceleration.

  (iii) Calculate the average resultant force on the car during this acceleration.

(b) When the car is travelling at its top speed of 350 km/h, the engine is producing its maximum power of 750 kW. Calculate the total resistive force (aerodynamic drag and friction) acting on the car at this speed.

(c) The car negotiates a flat, curved section of the track with a radius of 80 m at a constant speed of 180 km/h.

  (i) Calculate the centripetal force required to keep the car on this circular path.

(ii) Calculate the total downward normal force exerted by the track on the tyres.

(iii) Calculate the maximum possible frictional force between the tyres and the track.

(iv) By comparing your answers, determine whether it is possible for the car to make this turn.

(d) The MGU-K is used during a braking event from 300 km/h to 100 km/h.

(i) Calculate the total loss in kinetic energy of the car during this event.

(ii) Calculate the amount of electrical energy (in joules) recovered and stored in the battery.

(iii) Calculate the average electrical power generated by the MGU-K during this braking time.

(e) A spectator stands by the track. The car approaches and passes them at 300 km/h.

(i) Calculate the frequency of the sound the spectator hears as the car is approaching.

(ii) Calculate the frequency the spectator hears as the car is moving away.

Solutions

(a) (i) $v = 100 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 27.78 \text{ m s}^{-1}$.

(ii) $a = \frac{v-u}{t} = \frac{27.78-0}{2.6} = 10.68 \text{ m s}^{-2} \approx 10.7 \text{ m s}^{-2}$.

(iii) $F_{net} = ma = (800)(10.68) = 8544 \text{ N} \approx 8540 \text{ N}$.

(b) $v_{top} = 350 \text{ km/h} \times \frac{1000}{3600} = 97.22 \text{ m s}^{-1}$.
At constant top speed, the engine's thrust force equals the resistive force.
Using $P = Fv$:
$F_{resist} = \frac{P_{max}}{v_{top}} = \frac{750\times10^3 \text{ W}}{97.22 \text{ m s}^{-1}} = 7715 \text{ N} \approx 7720 \text{ N}$.

(c) (i) $v_c = 180 \text{ km/h} \times \frac{1000}{3600} = 50.0 \text{ m s}^{-1}$.
$F_c = \frac{mv^2}{r} = \frac{(800)(50.0)^2}{80} = 25000 \text{ N}$.

(ii) The total normal force N is the sum of the car's weight and the aerodynamic downforce.
$W = mg = (800)(9.81) = 7848 \text{ N}$.
$N = W + F_{down} = 7848 + 15000 = 22848 \text{ N}$.

(iii) $F_{fric,max} = \mu_s N = (1.5)(22848) = 34272 \text{ N}$.

(iv) The required centripetal force is 25000 N. The maximum available frictional force is 34272 N. Since $F_{fric,max} > F_c$, the car can successfully make the turn.

(d) (i) $v_1 = 300 \text{ km/h} = 83.33 \text{ m s}^{-1}$. $v_2 = 100 \text{ km/h} = 27.78 \text{ m s}^{-1}$.
$\Delta KE = \frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2 = \frac{1}{2}(800)(83.33^2 - 27.78^2)$
$\Delta KE = 400(6944 - 771.7) = 2.469 \times 10^6 \text{ J} \approx 2.47 \times 10^6 \text{ J}$.

(ii) $E_{recov} = \Delta KE \times \eta_K = (2.469 \times 10^6) \times 0.60 = 1.48 \times 10^6 \text{ J}$.

(iii) $P_{charge} = \frac{E_{recov}}{t_{brake}} = \frac{1.48\times10^6 \text{ J}}{2.0 \text{ s}} = 740000 \text{ W} = 740 \text{ kW}$.

(e) $v_{source} = 300 \text{ km/h} = 83.33 \text{ m s}^{-1}$.

(i) Approaching: $f' = f_s \left(\frac{v_{sound}}{v_{sound}-v_{source}}\right)$
$f' = 800 \left(\frac{343}{343-83.33}\right) = 800 \left(\frac{343}{259.67}\right) = 1057 \text{ Hz} \approx 1060 \text{ Hz}$.

(ii) Moving away: $f' = f_s \left(\frac{v_{sound}}{v_{sound}+v_{source}}\right)$
$f' = 800 \left(\frac{343}{343+83.33}\right) = 800 \left(\frac{343}{426.33}\right) = 644 \text{ Hz}$.

[Diagram of F1 car in a turn, showing $\text{F}_c, F_down, W, andF_friction$]

Figure 1: Forces on an F1 Car During Cornering

Cambridge International AS & A Level Physics (9702)

Paper 4: A Level Structured Questions

Variant 2 (Higher Difficulty) — Additional Open-Ended Section

Instructions to candidates

- Answer all questions in this section.
- Working should be shown clearly.
- Where appropriate, use SI units and give answers to a sensible number of significant figures.

Total for this section: 40 marks

Questions (Variant 2 only)

1 Quantum and thermal physics in a stellar gas [13]

A diffuse hydrogen cloud in a star-forming region has number density

$$n = 3.0 \times 10^{32}\,\mathrm{m}^{-3}.$$

Assume all particles are protons of mass $m_p = 1.67 \times 10^{-27}\,\mathrm{kg}$.

(a) Show that the Fermi energy of this proton gas is of order $10^{-18}\,\mathrm{J}$. Use

$$E_F = \frac{h^2}{8m_p}\left(\frac{3n}{\pi}\right)^{2/3}.$$

[4]

(b) Hence estimate the Fermi temperature T_F, where $E_F = kT_F$. [2]

(c) The actual cloud temperature is $T = 1.5 \times 10^7\,\mathrm{K}$. Discuss, with quantitative support, whether classical Maxwell–Boltzmann statistics or quantum degeneracy is the better model for the translational motion of the protons. [3]

(d) Open-ended: in no more than 6 lines, explain one astrophysical consequence if the density were increased by several orders of magnitude while temperature stayed similar. Your answer should link microscopic statistics to a macroscopic observable. [4]

2 Electric, magnetic and gravitational fields synthesis [14]

An ion beam source emits singly ionised neon ions $^{20}\mathrm{Ne}^+$ (mass $m = 3.32 \times 10^{-26}\,\mathrm{kg}$, charge $q = +e$). Ions are accelerated through a potential difference V, then enter a region of uniform magnetic field $B = 0.42\,\mathrm{T}$, normal to the velocity.

(a) Derive an expression for the circular path radius r in terms of m, q, B and V. [3]

(b) Calculate r for $V = 4.0 \times 10^3\,\mathrm{V}$. [2]

(c) The apparatus is moved to a low-Earth-orbit laboratory where a weak residual gravitational field causes a vertical deflection during the semicircular flight. For semicircle transit time, estimate the additional downward displacement y if local $g = 8.7\,\mathrm{m\ s^{-2}}$. State clearly any assumptions. [4]

(d) Open-ended: suggest and justify one practical design change to reduce uncertainty in measured m/q ratio below 0.5%. Your answer should identify the dominant source of uncertainty and explain how your modification addresses it. [5]

3 Nuclear decay and medical imaging design [13]

A short-lived radioisotope is considered for tracer imaging. The isotope decays with half-life $t_{1/2} = 18.0\,\mathrm{h}$, emitting one gamma photon per decay.

(a) Determine the decay constant λ. [2]

(b) A sample is prepared with initial activity $A_0 = 9.6 \times 10^8\,\mathrm{Bq}$. Calculate its activity after 30.0 h. [3]

(c) The detector records only 22% of emitted photons because of geometric and absorption losses. Find the expected count rate at 30.0 h. [2]

(d) Open-ended: discuss two competing constraints when choosing half-life for a diagnostic tracer (image quality, patient dose, and logistics of transport/use). Support your points with physics reasoning rather than general statements. [6]

Sample solutions (Variant 2)

1 Quantum and thermal physics

(a)

$$E_F = \frac{h^2}{8m_p}\left(\frac{3n}{\pi}\right)^{2/3} = \frac{(6.63\times10^{-34})^2}{8(1.67\times10^{-27})}\left(\frac{3(3.0\times10^{32})}{\pi}\right)^{2/3}.$$

Numerically,

$$E_F \approx 1.5\times10^{-18}\,\mathrm{J} \quad (\text{order } 10^{-18}\,\mathrm{J}).$$

(b)

$$T_F = \frac{E_F}{k} = \frac{1.5\times10^{-18}}{1.38\times10^{-23}} \approx 1.1\times10^5\,\mathrm{K}.$$

(c)

$$\frac{T}{T_F} \approx \frac{1.5\times10^7}{1.1\times10^5} \approx 1.4\times10^2 \gg 1.$$

Since $T \gg T_F$, thermal energy scale greatly exceeds degeneracy scale, so Maxwell–Boltzmann statistics is a good approximation for proton translational states; degeneracy effects are negligible.

(d) Possible high-level answer: increasing density raises $E_F \propto n^{2/3}$, so degeneracy pressure grows even without temperature increase. At sufficiently high density, this pressure becomes dynamically important and can oppose gravitational compression, altering stellar structure (e.g. compact-object behaviour, reduced compressibility, changed mass–radius relation).

2 Fields synthesis

(a) From acceleration,

$$qV = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2qV}{m}}.$$

In magnetic field,

$$qvB = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB} = \frac{1}{B}\sqrt{\frac{2mV}{q}}.$$

(b)

$$r = \frac{1}{0.42}\sqrt{\frac{2(3.32\times10^{-26})(4.0\times10^3)}{1.60\times10^{-19}}} \approx 9.7\times10^{-2}\,\mathrm{m}.$$

(c) Speed:

$$v = \sqrt{\frac{2qV}{m}} \approx 1.96\times10^5\,\mathrm{m\ s^{-1}}.$$

Semicircle arc length $= \pi r$, so

$$t = \frac{\pi r}{v} \approx \frac{\pi(9.7\times10^{-2})}{1.96\times10^5} \approx 1.56\times10^{-6}\,\mathrm{s}.$$

Assuming constant g, negligible effect on speed and magnetic force direction,

$$y = \frac{1}{2}gt^2 = \frac{1}{2}(8.7)(1.56\times10^{-6})^2 \approx 1.1\times10^{-11}\,\mathrm{m}.$$

(d) One valid proposal: increase uniformity and calibration of B using NMR Hall-probe mapping and active feedback coils. Since $m/q \propto B^2 r^2/V$, systematic error in B propagates strongly; reducing field non-uniformity and drift directly lowers bias in inferred m/q. Complement with longer path radius readout (larger r) to reduce fractional position-reading error.

3 Nuclear decay and imaging

(a)

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.693}{18.0 \times 3600} \approx 1.07 \times 10^{-5}\,\mathrm{s}^{-1}.$$

(b)

$$A = A_0 e^{-\lambda t} = 9.6 \times 10^8 \exp\left[-\left(1.07 \times 10^{-5}\right)(30.0 \times 3600)\right] \approx 3.03 \times 10^8\,\mathrm{Bq}.$$

(c) Detected count rate

$$R = 0.22A \approx 0.22(3.03 \times 10^8) \approx 6.7 \times 10^7\,\mathrm{s}^{-1}.$$

(d) Indicative discussion points:

- Short half-life reduces total dose after imaging (faster decay), but if too short, activity falls during preparation/transport and count statistics worsen.
- Long half-life eases transport and allows longer imaging window, but increases residual activity in patient and staff environment.
- Optimal choice balances sufficient photons during scan (signal-to-noise) against minimising integrated decay energy delivered to tissue.

Mark scheme (CIE 9702 P4 style, Variant 2 only)

Key

- **B** mark: independent factual statement
- **M** mark: method mark
- **A** mark: accuracy mark (dependent on valid method)
- **C** mark: communication / logical physics argument

Question 1 [13]

Part	Marking points	Marks
(a)	Substitution into correct Fermi-energy expression (M1); powers of ten handled correctly to obtain value of order 10^{-18} J (A1); reasonable numerical result around 1 to 2×10^{-18} J (A1); clear unit J (B1).	4
(b)	Uses $T_F = E_F/k$ (M1); correct numerical estimate $\sim 10^5$ K with unit (A1).	2
(c)	Computes or states comparison $T \gg T_F$ quantitatively (M1); identifies MB/classical as appropriate model (A1); links conclusion to negligible degeneracy effects (C1).	3
(d)	Any two valid linked ideas, e.g. increased $n \Rightarrow E_F \uparrow$, degeneracy pressure significance, macroscopic structural effect with explicit micro-to-macro link. Award up to 4 as C1+C1+B1+B1.	4

Question 2 [14]

Part	Marking points	Marks
(a)	Uses $qV = \frac{1}{2}mv^2$ (M1); uses magnetic force as centripetal (M1); derives $r = \frac{1}{B}\sqrt{\frac{2mV}{q}}$ or equivalent (A1).	3
(b)	Correct substitution (M1); correct radius $\approx 9.7 \times 10^{-2}$ m (A1).	2
(c)	Finds/uses transit time for semicircle (M1); applies $y = \frac{1}{2}gt^2$ (M1); correct tiny displacement order 10^{-11} m (A1); states valid assumptions (C1).	4
(d)	Identifies dominant uncertainty source (B1); proposes realistic design modification (B1); physics justification linking to m/q relation (C1); explains expected uncertainty reduction mechanism (C1); coherent technical communication (C1).	5

Question 3 [13]

Part	Marking points	Marks
(a)	Uses $\lambda = \ln 2/t_{1/2}$ with conversion of hours to seconds (M1); correct $\lambda \approx 1.07 \times 10^{-5}\,\mathrm{s}^{-1}$ (A1).	2
(b)	Uses exponential decay law correctly (M1); substitution with consistent units (M1); correct activity $\approx 3.0 \times 10^{8}\,\mathrm{Bq}$ (A1).	3
(c)	Applies detection fraction to activity (M1); correct count rate $\approx 6.7 \times 10^{7}\,\mathrm{s}^{-1}$ (A1).	2
(d)	Any two developed constraints with physics basis. Award up to 6 as C1+C1+C1+B1+B1+B1 for balanced discussion including image statistics, dose and operational timescale.	6

Total: 40 marks